QuickSight Configuration Reference — Robo Advisor Project

Dashboards were built in the QuickSight console, not from code files. This document records the parameter definitions, calculated-field expressions, and visual filter settings so the two analyses can be reproduced or handed off.

Database: `robo_advisor_tt` (Athena). All datasets connect via the Athena data source over the curated tables.

Analysis 1 — Robo Advisor (risk-tier recommendation dashboard)

Datasets used

- `fact_tier_weights` — allocation stacked bar (long form: rebal_date, tier_key, tier_name, ticker, weight)
- `fact_backtest` — growth-of-$1 line chart and max-drawdown bar (daily equity curves per tier + SPY)
- `fact_metrics` — risk-adjusted metrics table (one summary row per tier + SPY)
- `dim_user_profile` — persona profile table

Parameter

Property	Value
Name	`SelectedTier`
Data type	String
Values	Single value
Default value	`moderate`

One parameter, shared across datasets. A dropdown **control** is bound to it, so picking a tier sets `SelectedTier`.

Calculated field — benchmark preservation

The design goal: when a user filters to one tier, the SPY benchmark line/row must remain visible alongside it (never SPY alone, never the tier without SPY). This is done with a `keep_row` calculated field on each comparison dataset, then a visual filter keeping only `keep_row = 'show'`.

Critical detail: the SPY label differs between the two datasets, so the two `keep_row` formulas are NOT identical.

On `fact_backtest` (SPY labeled `SPY_benchmark` — underscore, no parens):

```
ifelse(
  {tier_name} = ${SelectedTier} OR {tier_name} = 'SPY_benchmark',
  'show',
  'hide'
)
```

On `fact_metrics` (SPY labeled `SPY (benchmark)` — space and parentheses):

```
ifelse(
  {tier_name} = ${SelectedTier} OR {tier_name} = 'SPY (benchmark)',
  'show',
  'hide'
)
```

Same field name (`keep_row`) on both — fine, since they live on different datasets. Only the SPY string literal changes.

Visual filter settings

Visual	Dataset	Filter
Growth of $1 (line)	`fact_backtest`	`keep_row` = include only `'show'`, scope: this visual
Risk-adjusted metrics (table)	`fact_metrics`	`keep_row` = include only `'show'`, scope: this visual
Tier allocation (stacked bar)	`fact_tier_weights`	plain filter on `tier_name` = `${SelectedTier}` (no SPY to preserve — this table has no SPY row)
Max drawdown (bar)	`fact_backtest`	`keep_row` = include only `'show'` (same pattern as growth)
Persona profile (table)	`dim_user_profile`	as needed per persona display

Expected behavior with default `moderate`: growth chart shows exactly two lines (moderate + SPY_benchmark); metrics table shows two rows (moderate + SPY (benchmark)). That two-series result is the proof the parameter → calculated field → filter chain is wired correctly.

Analysis 2 — Stock Exploration (S&P 500)

Standalone analysis. Single dataset.

Dataset used

- `fact_stock_screener` — one row per ticker (~503 rows), carrying YTD return, gainer/loser ranks, annualized volatility, max drawdown, 52-week high/low proximity, average dollar volume, and fundamentals joined from `dim_security`.

Visuals and settings

Visual	Type	Filter / setup
Top-10 YTD gainers	Horizontal bar	filter `rank_gainer <= 10`, sort ytd_return descending
Top-10 YTD losers	Horizontal bar	filter `rank_loser <= 10`, sort ytd_return ascending
Sector risk-return	Scatter (11 dots, one per GICS sector)	X = average `annual_vol` (formatted %), Y = average `ytd_return` (formatted %), tooltip = sector name + count

No parameter or calculated field required for this analysis — the ranks and metrics are precomputed in the `fact_stock_screener` CTAS, so the visuals filter and aggregate directly.

Notes for reproduction

- Both analyses read curated Athena tables; no direct raw-zone access.
- If a curated table is rebuilt, refresh the corresponding QuickSight dataset (SPICE refresh if using SPICE rather than direct query) for the dashboard to reflect new data.
- The `SelectedTier` parameter and the two `keep_row` calculated fields are the only non-trivial console configuration; everything else is standard field-to-visual assignment.